Courtroom Manager App

Mobile Case Management System for Courts

Project Overview

Courtroom Manager is a comprehensive Android application built with Flutter and Dart that streamlines courtroom operations by providing real-time case management for lawyers and court administrators. The app connects legal professionals with live court schedules, enabling efficient case tracking and timely notifications.

Tech Stack

- **Frontend:** Flutter & Dart
- **Backend/Database:** Firebase Realtime Database
- **Authentication:** Firebase Authentication
- **Push Notifications:** Firebase Cloud Messaging (FCM)
- **Development Environment:** Android Studio

Key Features

Dual User Profiles

1. **Lawyer Profile** - Track cases across multiple judges
2. **Case Manager Profile** - Real-time case order management

Core Functionalities

Real-time case updates across all users
Smart notifications when cases enter top 10
Drag-and-drop case reordering
Multi-judge case tracking
Case type categorization (Regular/Supplementary)
Paginated case listings (10 cases per page)
Visual highlighting for lawyer's own cases
Secure role-based authentication

Authentication System

The app features a **dual-role authentication** system with credentials stored in Firebase Realtime Database.

Login Interface

- Role selection dropdown (Lawyer/Manager)
- ID-based authentication
- Clean, intuitive material design

Security Features

- Firebase-backed authentication
- Role-based access control
- Secure session management

Lawyer Dashboard

Home Screen

Lawyers access two primary functions:

1. **Judges** - Browse all available judges and their case schedules
2. **Your Cases** - View all personally assigned cases

Judge Selection

- List of all judges with courtroom information
- Example judges:
    - Judge Robert Williams - Court Room 101
    - Judge Sarah Martinez - Court Room 202
    - Judge Michael Chen - Court Room 303

Case Viewing by Judge

When a lawyer selects a judge, they see:

- All cases scheduled for that judge
- Cases sorted by order/position
- **Visual Distinction:** Lawyer's own cases highlighted in amber/yellow
- Other lawyers' cases displayed in white
- **Pagination:** 10 cases per page with navigation controls

Case Information Displayed:

- Case title
- Case ID
- Lawyer name
- Case type badge (REGULAR or SUPPLEMENTARY)
- "DONE" status indicator
- Position in queue

Your Cases View

Comprehensive overview of all cases assigned to the logged-in lawyer across all judges.

Features:

- **Expandable Cards:** Tap to view detailed information
- **Status Indicators:**
  - ✓ Checkmark for completed cases
  - Green badge showing "DONE" status
  - Numbered position badges (1, 2, 3...)
- **Case Details:**
  - Case ID
  - Judge assignment
  - Current position in queue
  - Case type classification

Visual Organization:

- Completed cases shown with strikethrough text
- Active cases prominently displayed
- Position tracking (e.g., "Position: #1")

Screenshot Reference: "Your Cases" screen showing multiple cases including "Theft Charges - Martinez" at position #1

Smart Notification System

Real-time push notifications powered by Firebase Cloud Messaging.

Notification Triggers:

Lawyers receive notifications when their case:

- Enters the top 10 in any judge's queue
- Changes position significantly
- Receives status updates

Notification Details Include:

- Case title
- Current position (e.g., "17th in the courtroom")
- Judge name
- Timestamp

Example Notifications:

- "Your case Perjury Charges - Court Testimony is now 17th in the courtroom of Judge Sarah Martinez"
- "Your case Vandalism Charges - Public Property is now 15th..."
- "Your case Kidnapping Charges is now 11th..."

Case Manager Dashboard

Manager Responsibilities:

Each manager is assigned to a specific judge and controls their entire case docket.

Real-Time Case Management:

Add New Cases

Managers can add cases through an intuitive dialog:

- **Case ID** - Unique identifier
- **Case Title** - Full case name
- **Lawyer Selection** - Dropdown of available lawyers
- **Case Type** - Regular or Supplementary
- **Expected Time** - Scheduled hearing time

Case List Management:

- **Drag-and-Drop Reordering** - Simply drag cases to reorder priority
- **Delete Cases** - Red trash icon for removal
- **Status Marking** - Green checkmark to mark as "DONE"
- **Visual Feedback** - Position numbers update instantly

Current Case Display:

Each case card shows:

- Position number (in circle badge)
- Case title
- Case type badge (colored)
- Case ID
- Lawyer name
- Delete button

Screenshot References:

- Add Case dialog with all input fields
- Manager dashboard showing case list with drag handles and delete buttons

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Firebase Database Architecture

Database Structure:

```
courtroom_manager/
├── lawyers/
│   ├── LAW001
│   │   ├── name: "John Anderson"
│   │   ├── email: "john@example.com"
│   │   └── fcmToken: "..."
│   ├── LAW002
│   └── LAW003
│
├── managers/
```

```
│   ├── MGR001
│   │   ├── name: "Manager One"
│   │   └── assignedJudgeId: "JDG001"
│   └── MGR002
│
├── judges/
│   ├── JDG001
│   │   ├── name: "Judge Robert Williams"
│   │   └── courtRoom: "Court Room 101"
│   ├── JDG002
│   └── JDG003
│
└── cases/
    ├── case_001
    │   ├── caseId: "case_001"
    │   ├── caseTitle: "Divorce Settlement - Miller"
    │   ├── lawyerId: "LAW001"
    │   ├── judgeId: "JDG001"
    │   ├── caseType: "regular"
    │   ├── orderIndex: 0
    │   ├── expectedTime: "2026-01-16T10:00:00"
    │   └── lawyerName: "Emily Thompson"
    └── ...
```

Firebase Services Used:

1. **Realtime Database** - Instant synchronization across all devices
2. **Cloud Messaging** - Push notifications
3. **Authentication** - Secure user login

Data Flow:

1. Manager updates case order → Firebase Realtime Database
2. Database triggers real-time listeners in all connected apps
3. Lawyer apps instantly update case positions
4. If case enters top 10 → FCM sends notification
5. All changes persist and sync across sessions

UI/UX Design Highlights

Material Design Implementation:

- Clean, professional interface suitable for legal professionals
- Consistent color scheme (blue-grey primary palette)
- Card-based layouts for information hierarchy
- Intuitive icons and visual indicators

Visual Indicators:

- **Amber/Yellow Cards** - User's own cases
- **Green Badges** - Completed ("DONE") status
- **Blue Badges** - Regular cases
- **Purple Badges** - Supplementary cases
- **Numbered Circles** - Case position in queue

Responsive Design:

- Optimized for Android devices
- Scrollable lists for extensive data
- Dialog overlays for input forms
- Paginated views prevent information overload

Technical Implementation Highlights

Real-Time Synchronization:

- Firebase listeners on case collection
- Automatic UI updates on data changes
- No manual refresh needed

Efficient Data Loading:

- Paginated case lists (10 per page)
- Lazy loading for optimal performance
- Indexed queries for fast retrieval

Notification Logic:

- Background monitoring of case positions
- Position threshold detection (top 10)
- Targeted FCM token-based delivery

State Management:

- StatefulWidgets for dynamic UI
- Real-time data streams
- Automatic reordering on drag-and-drop

Use Case Scenarios

Scenario 1: Lawyer Checks Case Status

1. Lawyer logs in with ID (e.g., LAW001)
2. Selects "Your Cases" to see all assignments
3. Sees "Criminal Defense - Roberts" at position #1
4. Receives notification when position changes

Scenario 2: Manager Reorders Cases

1. Manager logs in with ID (e.g., MGR001)
2. Views all cases for assigned judge
3. Drags "Divorce Settlement - Miller" to top position
4. All lawyers viewing that judge see instant update
5. Lawyer for that case receives notification

Scenario 3: Adding New Case

1. Manager clicks "Add Case" button
2. Fills in case details in dialog
3. Selects lawyer from dropdown
4. Sets case type and time
5. Case appears in list and lawyer's dashboard simultaneously

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Project Impact

Benefits for Courts:

- Reduced confusion about case order
- Real-time communication with lawyers
- Digital record of all case movements
- Improved courtroom efficiency

Benefits for Lawyers:

- No need to physically wait at courthouse
- Instant notifications of case status
- Multi-judge case tracking in one app
- Clear visibility of queue position

Benefits for Managers:

- Easy case reordering via drag-and-drop
- Quick case addition and removal
- Real-time control over court schedule
- Reduced manual coordination

Future Enhancements

Potential features for future iterations:

- **Multi-language support** for diverse jurisdictions
- **Calendar integration** for automated reminders
- **Video conferencing** for remote hearings
- **Document upload** for case files
- **Analytics dashboard** for court administrators
- **Case history** and audit trail
- **Multiple courtroom** management per judge
- **Lawyer-to-lawyer messaging** within app

Project Conclusion

Courtroom Manager demonstrates a comprehensive understanding of:

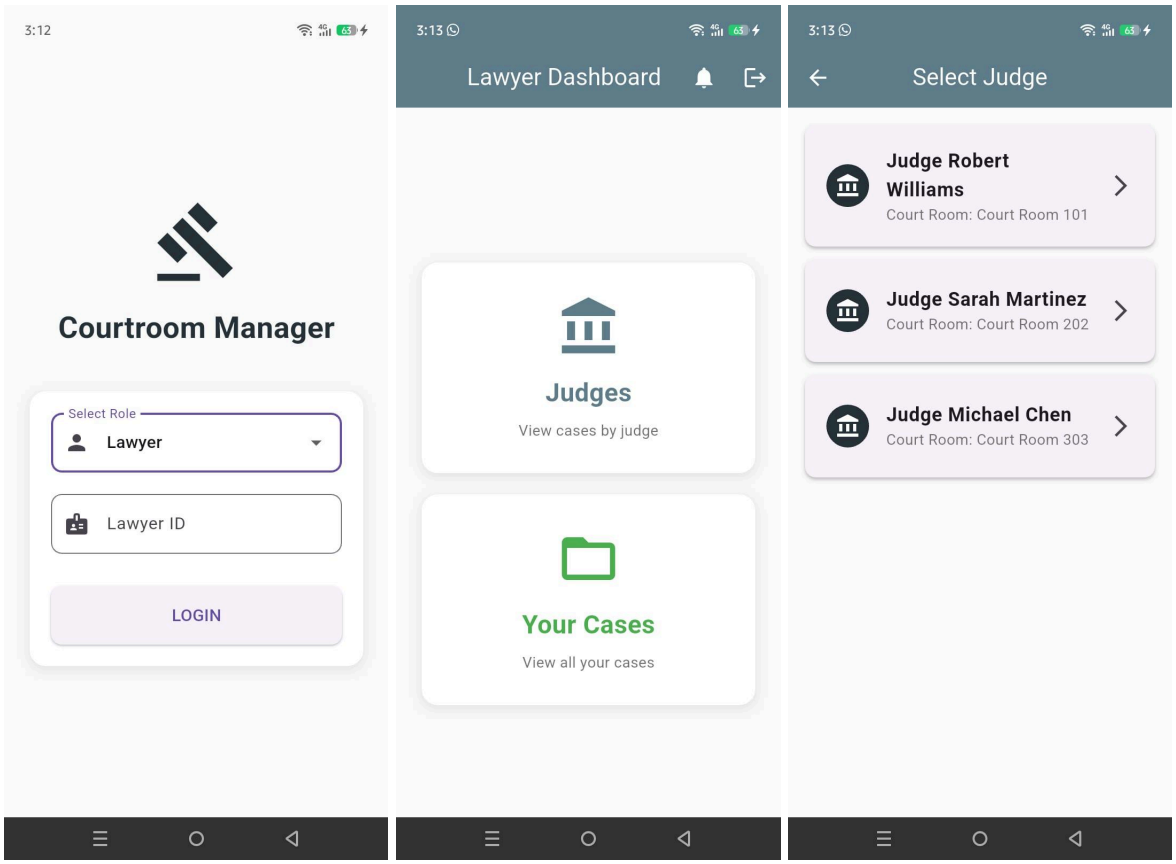
- Mobile app development with Flutter/Dart
- Cloud database integration (Firebase)
- Real-time data synchronization
- Push notification implementation
- Role-based authentication systems
- Material Design principles
- State management in Flutter
- RESTful architecture patterns

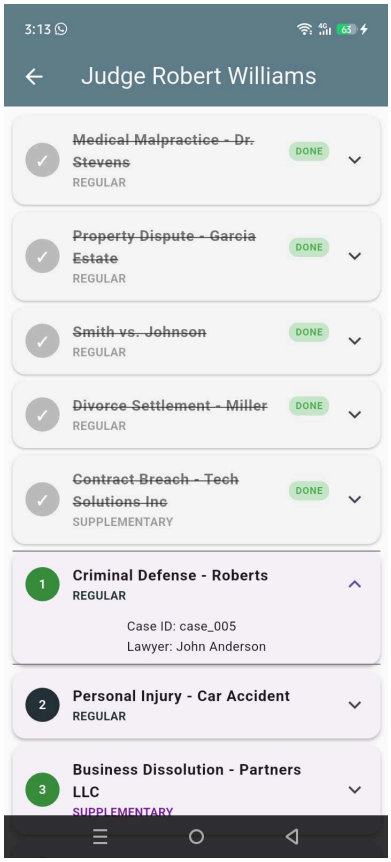
The app successfully solves real-world courtroom management challenges through an intuitive, efficient, and scalable mobile solution.

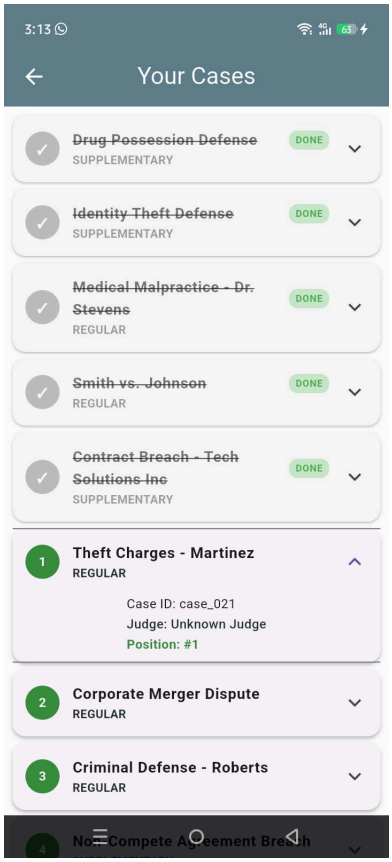
Technology Stack:

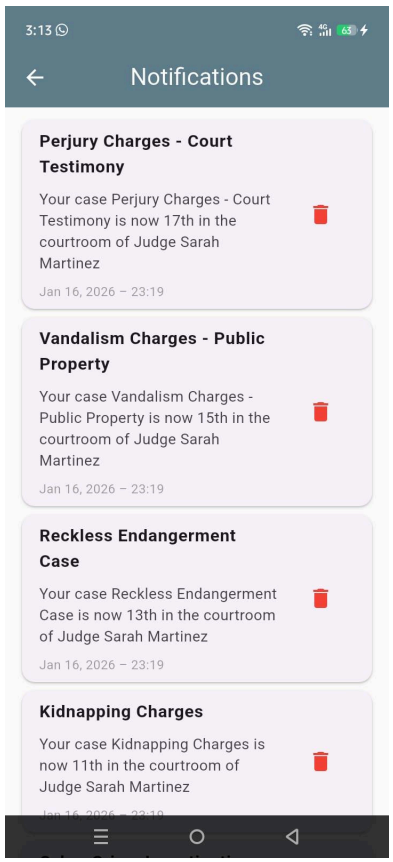
- Flutter 3.0+
- Dart 3.0+
- Firebase Realtime Database
- Firebase Cloud Messaging
- Android Studio

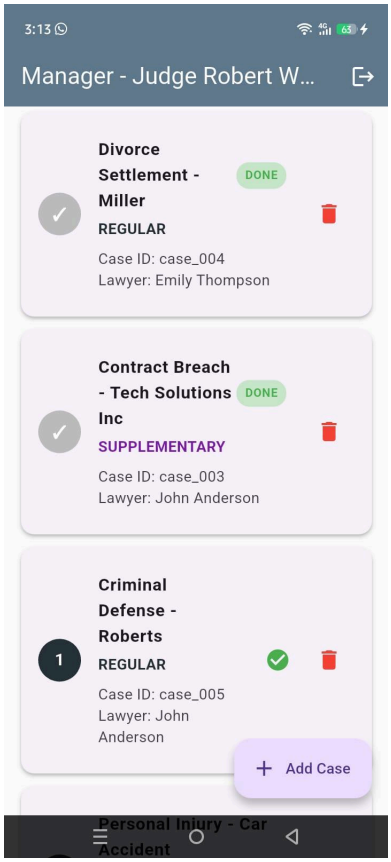
Some Screenshots:

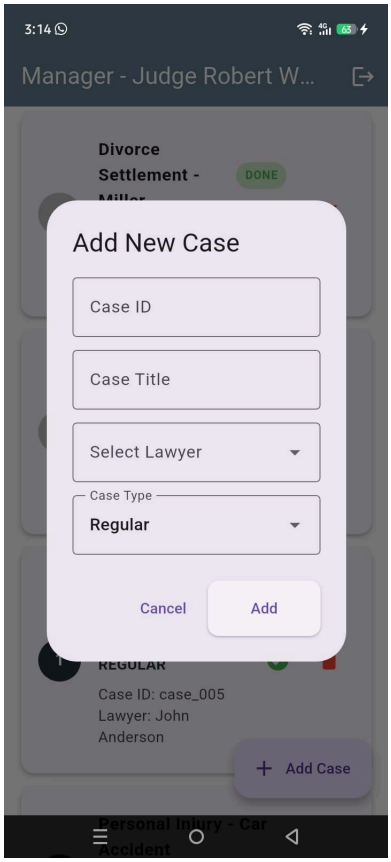



3:13
Judge Robert Williams
Medical Malpractice - Dr. Stevens
REGULAR
DONE
Property Dispute - Garcia Estate
REGULAR
DONE
Smith vs. Johnson
REGULAR
DONE
Divorce Settlement - Miller
REGULAR
DONE
Contract Breach - Tech Solutions Inc
SUPPLEMENTARY
DONE
1
Criminal Defense - Roberts
REGULAR
Case ID: case_005
Lawyer: John Anderson
2
Personal Injury - Car Accident
REGULAR
3
Business Dissolution - Partners LLC
SUPPLEMENTARY


3:13
Your Cases
Drug Possession Defense
SUPPLEMENTARY
DONE
Identity Theft Defense
SUPPLEMENTARY
DONE
Medical Malpractice - Dr. Stevens
REGULAR
DONE
Smith vs. Johnson
REGULAR
DONE
Contract Breach - Tech Solutions Inc
SUPPLEMENTARY
DONE
1
Theft Charges - Martinez
REGULAR
Case ID: case_021
Judge: Unknown Judge
Position: #1
2
Corporate Merger Dispute
REGULAR
3
Criminal Defense - Roberts
REGULAR


3:13
Notifications
Perjury Charges - Court Testimony
Your case Perjury Charges - Court Testimony is now 17th in the courtroom of Judge Sarah Martinez
Jan 16, 2026 – 23:19
Vandalism Charges - Public Property
Your case Vandalism Charges - Public Property is now 15th in the courtroom of Judge Sarah Martinez
Jan 16, 2026 – 23:19
Reckless Endangerment Case
Your case Reckless Endangerment Case is now 13th in the courtroom of Judge Sarah Martinez
Jan 16, 2026 – 23:19
Kidnapping Charges
Your case Kidnapping Charges is now 11th in the courtroom of Judge Sarah Martinez


3:13
Manager - Judge Robert W...
Divorce Settlement - Miller
DONE
REGULAR
Case ID: case_004
Lawyer: Emily Thompson
Contract Breach - Tech Solutions Inc
DONE
SUPPLEMENTARY
Case ID: case_003
Lawyer: John Anderson
1
Criminal Defense - Roberts
REGULAR
Case ID: case_005
Lawyer: John Anderson
Add Case
Personal Injury - Car Accident


3:14
Manager - Judge Robert W...
Divorce Settlement - Miller
DONE
Add New Case
Case ID
Case Title
Select Lawyer
Case Type
Regular
Cancel
Add
REGULAR
Case ID: case_005
Lawyer: John Anderson
Add Case
Personal Injury - Car Accident